

Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

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SUMMARY

Data scientist with a B.Sc. in Statistics focused on experiments and causal decisions. Ran an uplift study on 64,000 customers with S, T and X learners and a causal forest, built a power, MDE and CUPED toolkit, and delivered results as a two-page decision memo. Writes SQL and Python.

SKILLS

Statistics and Causal Inference: Experiment Design, Power Analysis, CUPED, Uplift Modeling, Causal Inference (S/T/X-Learners, Causal Forests), Qini and AUUC, Statistical Inference, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Data Analysis: SQL, Pandas, NumPy, Anomaly Detection

Engineering and MLOps: Python, Pytest, CI/CD (GitHub Actions), SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit, React

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Uplift Modelling for Campaign Targeting | [GitHub](#)

Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Showed the campaign works (1.25% conversion against 0.57%) but that a non-randomised list overstates the effect by 7.5×, and that a response model mails five times as many customers the campaign harms as an uplift model does.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.
- Added a power and MDE calculator and a CUPED variance-reduction demo; 135 tests on synthetic data with CI on Python 3.10 to 3.12.
- Hand-implemented S-, T- and X-learners over a swappable scikit-learn and LightGBM base plus an EconML causal forest, validated first on synthetic data with a known individual treatment effect.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.

Telco Customer Churn, End to End on AWS | [GitHub](#)

Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch

- Built a churn prediction model with feature engineering and an XGBoost classifier, with experiments tracked in MLflow.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.

- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER**

May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action